r/AskEngineers

If AI data centers are using up water should we be building them near polluted water so they can be used to as water distilleries?

Posted by u/sarcastagirly • • 0 points • 40 comments

Could data centers be built with "steam chambers" to contain the vapors and have the water put back into the ground supply?

Link about AI Data centers using so much water.
https://www.newsweek.com/why-ai-so-thirsty-data-centers-use-massive-amounts-water-1882374

Comments

u/Mango_Tree6316 • 1 points • 2024-11-08 20:19 UTC

i'm not understanding, is the water evaporating and being turned into steam? how do they use this much water? do they get that hot?

u/hughk • 5 points • 2024-07-18 11:36 UTC

The city of Frankfurt in Germany is a major hub for data centres in Europe due to international internet connectivity.

Water itself isn't so much consumed, just heated up. Many centres now use closed loop systems so after cooling, the water is reused but there is still the matter of dealing with the heat.

Recent data centres are plugged in to the district heating network. Germany forces that by law by 2030 but Frankfurt and the neighbouring town of Eschborn have already moved ahead.

I don't mention AI because that is just one specialised aspect of a data centre that provides compute power and online storage.

u/LordGarak • 3 points • 2024-07-18 09:18 UTC

The AI water problem is a short term issue. Right now there is a massive race to be first to market and profit from AI. So large companies are just throwing money at brute force AI training. As time goes on a few things may happen to alleviate the demand for electricity and cooling. The bubble may burst and investment will dry up. The processors will get way more efficient or atleast operate at more efficient power levels. Someone could win the race and most of the training data centers will shut down. The training process could get much less processing intensive(software improvement).

Right now it doesn't seem like anyone has really figured out how to capitalize on AI. They are all just brute forcing their way along to be best and attract investment. So this suggest that the bubble is going to burst and many of the AI datacenters may shut down.

AI isn't going to go away. Just the insane mad rush that is going on will die down. This will give time for all the hardware and software to get more efficient.

The water shortages are a local problem. There are many other places these datacenters could be built that have access to practically unlimited water. Electricity isn't as cheap, but trying to recapture water vapor would be way more expensive.

These places just need to charge more for the water. The datacenters would likely just move somewhere else if the cost of the water out weighed the savings of the cheap electricity.

u/getting_serious • 10 points • 2024-07-18 08:30 UTC

Good lord is that badly written. Ren's paper also seems a whole lot of words about little substance, and what the guy is quoted with in the article is actually manipulative. AI data centers need water?

No, they don't *need* water in the same way that hikers don't *need to* litter. It's just a convenience as long as the cost is zero.

That water usage has to be permitted. There also can be a price tag attached. As long as evaporating water is cheaper than running fans to blow air through cooling fins, it'll be done with water. If it's more expensive, the practice will stop at once.

There are zero reasons why AI semiconductors suddenly couldn't be cooled the same way that we've always cooled our computers.

u/NobodySpecific • 1 points • 2024-07-18 18:51 UTC

> As long as evaporating water is cheaper than running fans to blow air through cooling fins, it'll be done with water. If it's more expensive, the practice will stop at once.

So I am an engineer that develops both air and water cooled servers. Our water cooled servers absolutely do need to be water cooled in order to achieve the performance that customers demand. There is simply too much heat generated to effectively air cool it to a lower enough temperature to get the performance and reliability needed.

Our air cooled servers are generally much lower density (so more space to move air, fewer processors generating heat) and much lower performance.

u/getting_serious • 1 points • 2024-07-18 19:25 UTC

Cool! I have watched power density go up over the years, and I've heard people at an internal conference saying that their partners were shocked about > 50kW per rack.

But I am still skeptical regarding this article. Are you doing recirculating watercooling, or the kind of evaporative watercooling that Ren talks about? I don't mean quick disconnect, I mean actual "medium never comes back to us".

u/NobodySpecific • 1 points • 2024-07-18 19:31 UTC

> Are you doing recirculating watercooling, or the kind of evaporative watercooling that Ren talks about? I don't mean quick disconnect, I mean actual "medium never comes back to us"

My involvement ends with the processor and how it integrates into the drawer. The cooling techniques are beyond my expertise. I do know that there is EXTENSIVE work done in that space to get as much efficiency as possible.

For example, we just changed fan manufacturers because they could generate significantly more air flow for the same power. I also have seen (but don't remember) presentations about water cooling innovations.

u/getting_serious • 1 points • 2024-07-18 19:37 UTC

It's exciting stuff for sure. I've moved on since but I'm still on a CPU that I haven't exactly had to pay for, and I'm watercooling it just because I'd never done that before. It's a fun thing, but the fine-pitched fins required for decent Re numbers also make for an excellent filter. I can only imagine the dirt pileup if you don't recirculate.

u/evil_boy4life • 5 points • 2024-07-18 08:06 UTC

Ok, I might be wrong but can anyone explain to me how the f** you're going to produce steam when the operating temp of your processors (the hottest point in your cycle) is actually about 30 degrees below the boiling point of water.

u/ChasteScape • 0 points • 2024-07-18 17:40 UTC

Drop the pressure.

u/evil_boy4life • 2 points • 2024-07-18 20:09 UTC

Never thought I'd say this but no, we will not build datacenters at the top of mount Everest.

u/sabretooth_ninja • 0 points • 2024-07-18 23:58 UTC

Wouldnt need water cooling up there

u/Infamous_Advantage37 • 1 points • 2024-07-19 22:08 UTC

Incorrect. They almost definitely would.